Unit 1: Foundations of Artificial Intelligence
- Linear models for regression and predictive analytics
- Logistic regression and probabilistic classification models
- L1 and L2 penalty methods for feature selection and regularization

Unit 2: Tree-Based Learning and Ensembles
- Decision tree induction with entropy and Gini impurity
- Random forest ensembles and bagging strategies
- Extreme Gradient Boosting (XGBoost) and LightGBM

Unit 3: Deep Neural Architectures
- Multilayer perceptrons, backpropagation, and stochastic gradient descent
- Convolutional Neural Networks (CNNs) for computer vision
- Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM)
- Attention mechanisms and Transformer foundations

Unit 4: Clustering and Dimensionality Reduction
- K-means clustering algorithm and cluster validation indices
- Principal Component Analysis and singular value decomposition
- t-Distributed Stochastic Neighbor Embedding (t-SNE) for visualization

Unit 5: Responsible AI and Deployment
- Model explainability using SHAP and LIME
- Algorithmic fairness, bias detection, and ethics in AI systems
- MLOps pipeline, model monitoring, and drift detection